

Scaling immersive visualization with COTS VR: the Voxar Labs experience

João Marcelo Teixeira and Veronica Teichrieb
Voxar Labs - Centro de Informática
Universidade Federal de Pernambuco
Recife, Pernambuco, Brazil
jmxnt@cin.ufpe.br, vt@cin.ufpe.br

ABSTRACT

Over the past decade, new commercial off-the-shelf (COTS) virtual reality (VR) devices have substantially lowered the entry barriers for immersive visualization in academic and industrial settings. Voxar Labs, based at the Federal University of Pernambuco (UFPE), has consistently favored smaller, more flexible installations and HMD-based immersive solutions over large-scale, high-cost visualization environments. This paper presents Voxar Labs' strategies for fostering immersive research and innovation with limited resources, discusses the advantages and challenges of adopting COTS technology, and highlights successes in education, research, and outreach. Our experience suggests that nimble, decentralized, and portable approaches can broaden the impact of immersive visualization labs while still meeting the demanding requirements of scientific research.

Index Terms: Immersive Visualization, COTS VR Technology, Research and Innovation

1 INTRODUCTION

Large-scale immersive visualization laboratories—once prohibitively expensive—have witnessed strong competition from cost-effective commercial off-the-shelf (COTS) solutions such as standalone VR headsets. Over the past 30 years, many institutions worldwide invested in powerwalls, CAVEs, and other large installations, but the ongoing advancements in head-mounted displays (HMDs) and readily available motion tracking solutions motivate new ways to deliver immersive computing experiences at reduced costs. Consequently, one of the central questions in the immersive visualization community has become: *Do we still need large, permanent immersive labs, or can flexible, smaller-scale setups based on COTS devices deliver equivalent value to researchers, students, and industry partners?*

This paper addresses these questions from the perspective of *Voxar Labs* [8], a research group founded in 2011 at the Center of Informatics (CIn) of the Federal University of Pernambuco (UFPE), Recife, Brazil. Voxar Labs focuses on spatial computing, with an emphasis on extended reality (XR), computer vision, and natural interaction. Rather than maintain a single, large, centrally located immersive facility, the lab advocates for distributed or portable immersive solutions. In particular, Voxar Labs invests in COTS devices such as the Meta Quest line of VR headsets. This approach has proven successful in bridging research, industrial collaboration, and entrepreneurial initiatives while achieving a broad societal impact.

2 BACKGROUND

2.1 Voxar Labs Overview

Voxar Labs develops research, development, and innovation (R&D&I) in extended reality (AR, MR, VR), computer vision, and natural interaction. Since its founding, the group has published extensively, registered patents, produced multiple spin-off companies, and earned national and international awards (including best paper awards in key venues like SVR and IEEE ISMAR competitions). The lab comprises roughly 40 members, including faculty, senior researchers, postdoctoral fellows, and undergraduate and graduate students across several departments—Computer Science, Electronic Engineering, Biomedical Engineering, and Design.

From its inception, Voxar Labs has taken a problem-focused approach, aiming to build solutions relevant to society and industry. By actively collaborating with companies, government institutions, and other universities, the lab ensures that research outcomes can quickly transition into impactful products and services. This strategy has led to various success stories in areas such as 3D tracking algorithms for mobile AR, interactive physiotherapy solutions using VR, and collaborative robotics.

2.2 Avoiding Large-Scale Installations

Unlike many immersive laboratories that invest heavily in multi-projector CAVEs or powerwalls, Voxar Labs emphasizes distributed immersive work across *flexible, individual HMDs*. Early on, the lab adopted devices such as Microsoft HoloLens, HTC Vive, and Oculus Quest, integrating them into diverse research projects, undergraduate classes, and outreach events. The absence of a massive dedicated VR space allowed the group to remain agile and scale up (or down) deployments with minimal overhead.

Over the years, this approach has led to a variety of novel applications in academic and industrial contexts. By focusing on portability and user-centric design, the lab has helped local industries adopt immersive training, created solutions for remote physiotherapy, and leveraged VR for inclusive educational experiences.

3 APPROACH AND ACTIVITIES

3.1 Infrastructure and Hardware

The lab currently employs multiple Meta Quest headsets to support immersive experiences in teaching, research, and outreach. Beyond the convenience of all-in-one HMDs, Voxar Labs leverages 360° cameras to capture real-world environments for VR-based tourism and industrial training solutions. Proposals such as the *LIVRE – Laboratório de Ensino de Realidade Virtual Imersiva* aim to further expand hardware availability across campus, ensuring that students from Computer Science, Electronic Engineering, Design, and other fields can access VR technology in a portable and interdisciplinary manner.

Recent internal funding at UFPE supported the purchase of 10 Meta Quest 3 devices, which are shared among three different centers—CIn, the Department of Electronics and Systems (CTG),

and the Department of Design (CAC). This “mobile lab” arrangement meets the needs of multiple courses and student projects simultaneously, shifting the perspective of a dedicated immersive space toward one of location-agnostic, *itinerant* VR solutions.

3.2 Teaching and Research Integration

3.2.1 Undergraduate and Graduate Courses

Voxar Labs faculty embed VR technology into various courses:

- **Undergraduate Capstone Projects:** Students in Computer Science or Electronic Engineering design VR-based prototypes for topics such as robotics, industrial monitoring, or interactive visualization.
- **Design Coursework:** Graduate design students explore user experience (UX) and user interface (UI) factors in immersive technologies, emphasizing interdisciplinary collaboration with engineers and computer scientists.
- **Project-Based Learning (PBL):** The “Prototyping and Development” courses allow students to propose and implement VR solutions for real-world problems, reinforcing the lab’s problem-focused mission.

Several of these projects have culminated in peer-reviewed publications at prominent conferences (e.g., IEEE ISMAR, SVR, SBGames), best paper awards, and spin-off startups.

3.2.2 Research Collaborations

Voxar Labs has contributed to advanced VR/AR research through both national and international partnerships—e.g., with INRIA, TUGraz, Kyushu University, and Harvard GSE. Joint initiatives have produced 3D tracking breakthroughs, semantic modeling solutions, and remote physiotherapy platforms that harness COTS VR devices. Recently, a partnership with UNIOESTE and Itaipu Parquetec has yielded cutting-edge approaches for VR-based training and environmental simulations.

3.3 Outreach and Societal Impact

From the beginning, Voxar Labs pursued a vision of *impactful research*. Rather than confine solutions to specialized hardware labs, the group leverages the portability of HMDs to engage with public schools, community events, and hackathons. For instance, one extension project used 360° videos displayed on standalone headsets to bring immersive tours of UFPE’s facilities to students in remote regions of Pernambuco. Another collaboration with tourism students at UFPE explored novel VR-based cultural experiences, highlighting local heritage sites.

These outreach efforts illustrate how decentralized, low-cost VR devices serve as catalysts for democratizing immersive experiences, aligning with the lab’s commitment to education quality and sustainable innovation (related to the United Nations’ Sustainable Development Goals — particularly ODS 4 and 9).

In a review from prof. Mark Billinghurst in 2018 [1], he systematically searched Scopus papers to find the top 160 authors in Augmented Reality (AR) and the institutions that they come from (Figure 1). The list is based on the number of publications produced by each author. Veronica Teichrieb is the only researcher from Latin America to appear in that list. When using the same criteria for Latin America, 5 from the top 10 researchers belong to the Voxar Labs team.



Figure 1: Where in the world is AR Research Happening? Medium publication from prof. Mark Billinghurst in 2018 [1].

In 2020, Voxar Labs CIn-UFPE was proud to organize IEEE Symposium on Mixed and Augmented Reality (ISMAR) together with the three major Brazilian conferences in VR/AR, CG/CV and Digital Games (all online due to the pandemics).

4 RECENT PUBLICATIONS DERIVED FROM THE ACQUIRED META QUEST 3 DEVICES

Since the acquisition of ten Meta Quest 3 headsets in late 2023, several new lines of research have emerged at Voxar Labs. These studies demonstrate how cutting-edge VR hardware enables novel applications in education, heritage preservation, and clinical practice. In the subsections below, we summarize a few key publications from 2024 that exemplify the lab’s progress and interdisciplinary collaborations. These publications highlight the versatility and interdisciplinary nature of research enabled by the newly acquired Meta Quest 3 devices. From surgical training and architectural preservation to children’s well-being, positive psychotherapy, and accessibility solutions, the headsets have proven effective in bridging technical and societal challenges. Voxar Labs continues to refine and expand upon these findings, emphasizing broader collaborations, real-world testing, and a long-term vision of using portable VR to extend the boundaries of immersive visualization beyond traditional laboratory settings.

4.1 The Hand Guide: A Virtual Reality Application for Surgical Training

Surgical students must practice procedures in a controlled and safe to make mistakes environment before they perform these procedures on patients. Despite that, surgical practices in these conditions are traditionally limited by the tutor’s time availability to provide feedback and by the usage of consumable materials to simulate surgery in a patient. In this context, the “Hand Guide”, a virtual reality application (Figure 2) that allows expert surgeons to record their hand movements during a simulated surgical procedure and offer it to a surgical student to try to replicate it, was developed. The application provides real-time feedback and also a final score at the end of the recorded procedure. Validation with a specialist revealed that the application may be used to teach theoretical content to students with no surgical background and to demonstrate new techniques to experienced surgeons [6].

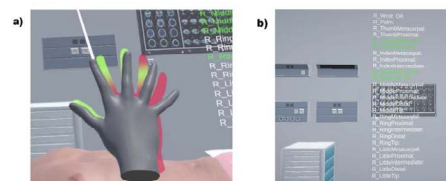


Figure 2: The visual (a) and textual (b) feedback provided by the application. Key points on a correct position have their name and vertices colored in green.

4.2 What if Video See-Through in HMDs Changes How Accurately We Perform Tasks?

This paper explores the impact of video see-through head-mounted display devices on user perception and accuracy during activity execution. Focusing on the Meta Quest 3 and employing two-stage experiments (dart throwing and bottle handling) (Figure 3), the study assesses both precision and everyday usability in mixed-reality tasks. Results provide empirical evidence on the feasibility of VST-HMDs for enhancing user interaction and underscore the need for further investigation into potential biases and perceptual shifts induced by see-through technologies [3].

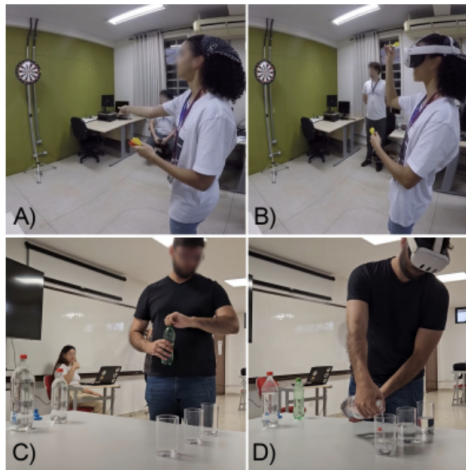


Figure 3: Darts without HMD (A) and with HMD (B); Bottle without HMD (C) and with HMD (D).

4.3 What if Virtual Reality Could Be Used as a Tool in Positive Psychotherapy in Brazil?

This article examines Virtual Reality (VR) as a tool in clinical psychology, specifically focusing on positive psychotherapy. The authors developed and tested VR scenarios (Figure 4) using the Meta Quest 3 that immersed patients in tailored environments to foster well-being. Interviews and preliminary trials with psychologists suggested that VR can improve therapeutic engagement, yet the authors highlight the importance of technical literacy among practitioners and patients. Future directions include refined narratives and validated protocols to enhance therapy outcomes [4].

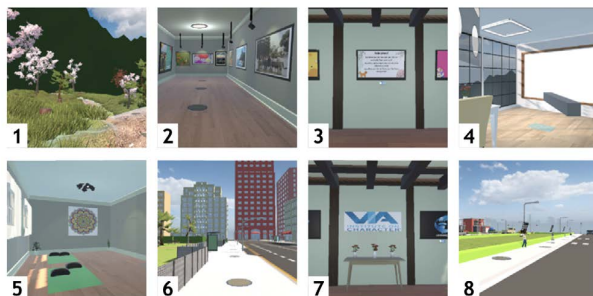


Figure 4: Skills - 1) Perceive positive events; 2) Amplifying positive events; 3) Gratitude; 4) Activation of positive emotions; 5) Mindfulness; 6) Positive reappraisal; 7) Personal strengths; 8) Acts of kindness

4.4 What if? We Could Virtually Restore and Visit a Pre-War Church to the Highest Fidelity Possible with No 3D Scanning Available?

Addressing the challenge of reconstructing historical sites destroyed or altered by warfare, this work integrates archival data, photogrammetry, and virtual modeling for the church of Thillois, heavily impacted by World War I. The Meta Quest 3 was employed to create immersive environments for users to explore a high-fidelity virtual recreation of the church (Figure 5). Despite limitations from unavailable 3D scans, the research demonstrates how VR can aid in architectural preservation, deepen historical engagement, and strengthen cultural memory [7].



Figure 5: Church of Thillois in its current state, the authors testing the Virtual Reality study at UFPE, tests conducted in the church of Thillois by residents. Possible idea for extension of the work using Mixed Reality with the insertion of the disappeared altarpiece in the current church.

4.5 What if... We Knew the Effects and Benefits of Virtual Reality in Childhood?

Offering a systematic review of Head-Mounted Display (HMD) VR solutions for children, this paper investigates health and safety issues, educational uses, and device suitability based on international studies. A total of 59 research articles were reviewed, indicating both positive outcomes—such as anxiety reduction and social skills improvement—and concerns related to prolonged exposure, such as physical discomfort and fatigue. The authors provide guidelines on safe usage times and best practices for HMD-based interventions with children, drawing on pilot studies conducted using Meta Quest 3 headsets (Figure 6) [2].

4.6 What if HMDs Could Generate Audio Descriptions of Real Scenes for Visually Impaired People?

This research explores real-time audio descriptions using AI-based computer vision on the Meta Quest 3. By capturing live video feeds and converting them into structured audio narratives, visually impaired users can navigate real-world environments more safely and confidently. Pilot studies conducted in indoor (Figure 7) and outdoor settings show promising accuracy in object recognition and spatial description, emphasizing the potential of HMD technology to expand accessibility and inclusivity [5].

5 LESSONS LEARNED

Voxar Labs' experience offers lessons relevant to other institutions evaluating their immersive infrastructure strategies:

1. **Cost and Scalability:** Relying primarily on COTS devices significantly reduces initial and maintenance costs compared to large-scale immersive laboratories. This approach allows incremental upgrades and expansions based on evolving research demands and market trends.

2. **Accessibility and Usability:** Portable HMDs increase student and faculty accessibility to immersive technology and ease the integration of VR into multiple courses. However, dedicated training and user support remain crucial to ensure broad adoption.
3. **Collaboration and Interdisciplinarity:** Mobile VR solutions facilitate cross-department projects—students from engineering, design, and other fields can share devices and produce comprehensive solutions. Key to success is a coherent scheduling and resource-sharing system.
4. **Sustainability:** Avoiding large, fixed installations aligns with sustainability goals by minimizing energy consumption, hardware footprint, and frequent expensive overhauls.
5. **Software Infrastructure:** While each project often develops custom solutions, Voxar Labs increasingly advocates open-source frameworks and standardized toolkits to streamline development and maintenance.

Benefits • Distraction, pain and anxiety reduction in critical environments • Development of social, motor, and communication skills • Learning in educational environments and language development • Testing of VR systems and serious games for various purposes	Difficulties • Device weight affecting comfort and usability • Lack of standardization regarding the studied age ranges • Limited usage time of the devices, lack of long-term indicators • Insufficient data on negative effects and overall impacts
Opportunities • Development of devices better suited for children • Long-term research to evaluate continuous impacts • Greater incorporation in educational contexts promoting engagement • Inclusion of more diverse populations and environments	Limitations • Lack of clear safety guidelines and regulations • Insufficient data on negative effects that may suggest underreporting of risks related to the use of HMD • Inconclusive results due to wide range of age groups, device types and exposure times

Figure 6: Summary of key findings from the review: benefits, difficulties, opportunities, and limitations.



Figure 7: Visually impaired person during experiments.

6 FUTURE DIRECTIONS

Moving forward, Voxar Labs aims to consolidate its “itinerant VR laboratory” model, expanding the repertoire of devices and software frameworks available to faculty and students. By establishing a robust scheduling system and continued collaborations with external partners, the lab envisions further enhancing interdisciplinary VR courses, applied research programs, and outreach initiatives.

Moreover, the lab’s trajectory may provide a blueprint for institutions seeking *lightweight and cost-effective* ways to engage with immersive computing. Beyond VR, Voxar Labs also invests in augmented reality (AR) and mixed reality (MR) solutions for multi-user, remote collaboration, and is exploring advanced haptic interfaces that remain compatible with existing HMD-based setups. Ultimately, the lab’s efforts aim to inspire a new generation of scalable, portable, and impact-driven immersive visualization laboratories.

7 CONCLUSION

Voxar Labs’ decision to adopt commercial VR headsets has proven effective for research, education, and outreach. The lab’s trajectory illustrates how flexibility, interdisciplinary collaboration, and a focus on practical applications can yield significant impact without the overhead of expensive, fixed CAVES or powerwalls. As COTS devices continue to improve, we anticipate that portable, distributed labs will increasingly satisfy the complex demands of immersive visualization, fostering inclusive participation, sustainability, and wide-ranging societal benefits.

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